Massachusetts Institute of Technology

Slow Control Update

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Slow Control

- Slow control stores experimental information that is not event-by-event
  - Currents, HV, magnetic field, etc.
- Updates if the change from previous value is over certain threshold
  - Often ~1-2 second level
- Stored as a TObject in ROOT file - usually
  - Was a bug with DAQ and not stored in most files near beginning of the beam time
  - Jan created new plugin to “regenerate” the object
  - DAQ was also fixed so slow control was saved as intended
- To keep things uniform, I modified the “regen” plugin to either actually regenerate the slow control or just copy it if it exists
- Ran over all runs so we have a complete set of slow control data in uniform format

Slow Control

- **Bottom line, how does one access and use slow control?**
- **Requires a recipe modification, which I have forced upon all of you**
- **YOU MUST PROVIDE THE SLOWCONTROL FILE IN YOUR DETECTOR COOKER CALLS**

```
[darklight]
   @include: init/all.ini
   source=slowctrl:DLMT
   destination=TrigScint

[plugins]
   slowctrl=libSlowCtrl
   TrigScint=libTrigScint

[startup]
   slowctrl=init
    TrigScint=startup

[execute]
   slowctrl=process
    TrigScint=process1

[postprocess]
    TrigScint=calib

[execute2]
   slowctrl=process
    TrigScint=process2

[finalize]
    TrigScint=finalize
```

Slow Control

```
cooker recipes/TrigScint/trs.ini /data/darklight/DLrun_002797.root run2797_trs.root
```

NO

Slow Control

```
cooker recipes/TrigScint/trs.ini /data/darklight/DLrun_002797.root GARBAGE.root
```

NO

Slow Control

```
cooker recipes/TrigScint/trs.ini /data/slowcontrol/run2797_slowcontrol.root:/data/darklight/DLrun_002797.root run2797_trs.root
```

YES!

Slow Control

```
cooker recipes/TrigScint/trs.ini /data/slowcontrol/run2797_slowcontrol.root /data/darklight/DLrun_002797.root run2797_trs.root
```

YES!

Slow Control

- The slow control generated files sit in /data/slowcontrol on the TRIUMF cluster
- You should never need to create or touch these files
- My code always works perfectly the first time, so there are no errors and it just works
    - Please direct problems to /dev/null, Laura, Win, etc.

Slow Control

- You only need to include the additional input file in detector level plugins.
- Higher level plugins can access it in the standard way
- Must include the “process” tags first, and for every execute pass

```
[darklight]
   @include: init/all.ini
   source=TrigScint:lumigemcooked
   destination=outputTracks

[plugins]
   slowctrl=libSlowCtrl
   GeomLoader=libGeomLoader
   GEMTrack=libGEMTrack

[startup]
   slowctrl=init
   GeomLoader=startup
   GEMTrack=startup

[execute]
   slowctrl=process
   GEMTrack=process

[finalize]
   GEMTrack=finalize
```

Slow Control

- **How does one actually access slow control in cooker?**
- **Modify recipe if needed**
- **Define your datum in your header**
- **Access in “startup”**
- **Use in code**

```
class simple_physics:public Plugin
{
 private:
    VertReconSet *vertices;
    slowctrl::datum * slowctrl_left_field;
    slowctrl::datum * slowctrl_right_field;
```

In the header

In startup

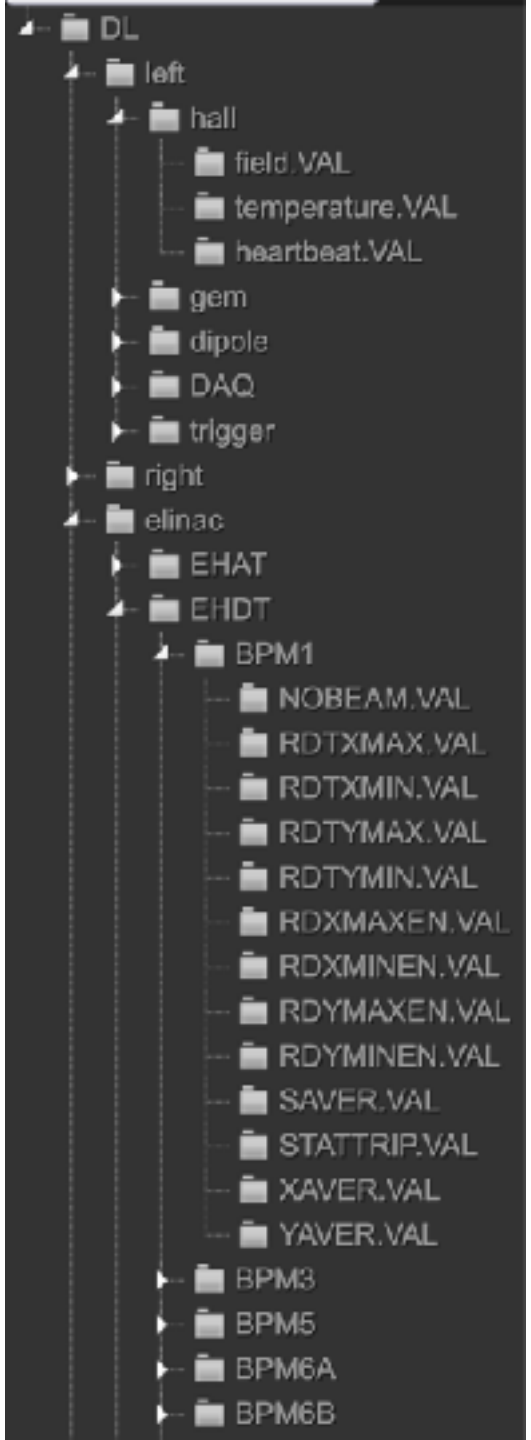
```
  //get the slowctrl manager
auto scmanager= (slowctrl::manager*) getMemoryObject("SlowCtrl Manager");
  if (!scmanager)
    {
      debug(0,"Couldn't find SlowCtrl manager. Plugin included?\n");
      exit(-1);
    }
  slowctrl_left_field=scmanager->getLastValidByName ("DL:left:hall:field.VAL");
  slowctrl_right_field=scmanager->getLastValidByName("DL:right:hall:field.VAL");
```

Use in code

```
mom=vertex.pred.rel_momentum*field_to_central*fabs(slowctrl_left_field->value);
```

Slow Control

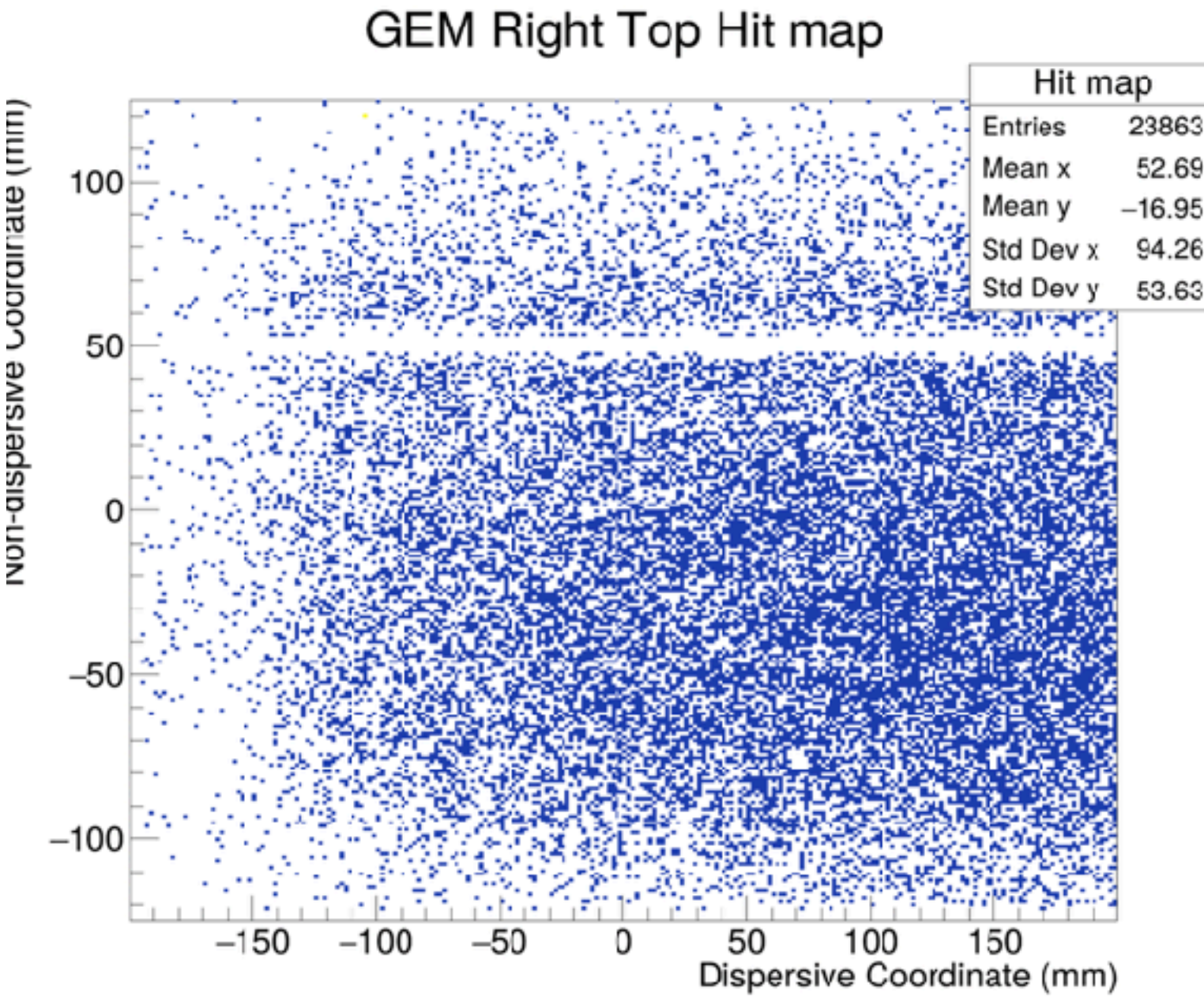
- What slow control variables exist and are usable?
- No one really knows
  - (Supposedly I do)
- Variables were added during the run
- Anything in the “timeline plot” section of the website existed at one point
- Separate the folders with a “:” in the call, so
  - DL:left:hall:field.VAL
  - DL:elinac:EHDT:BPM1:NOBEAM.VAL



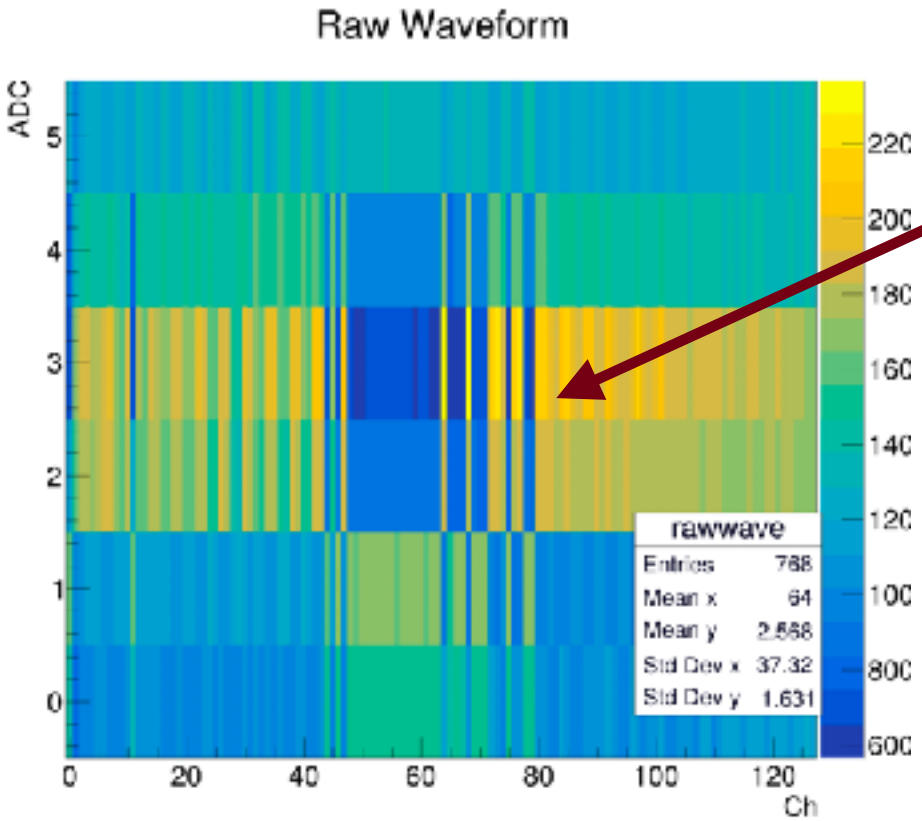
Slow Control

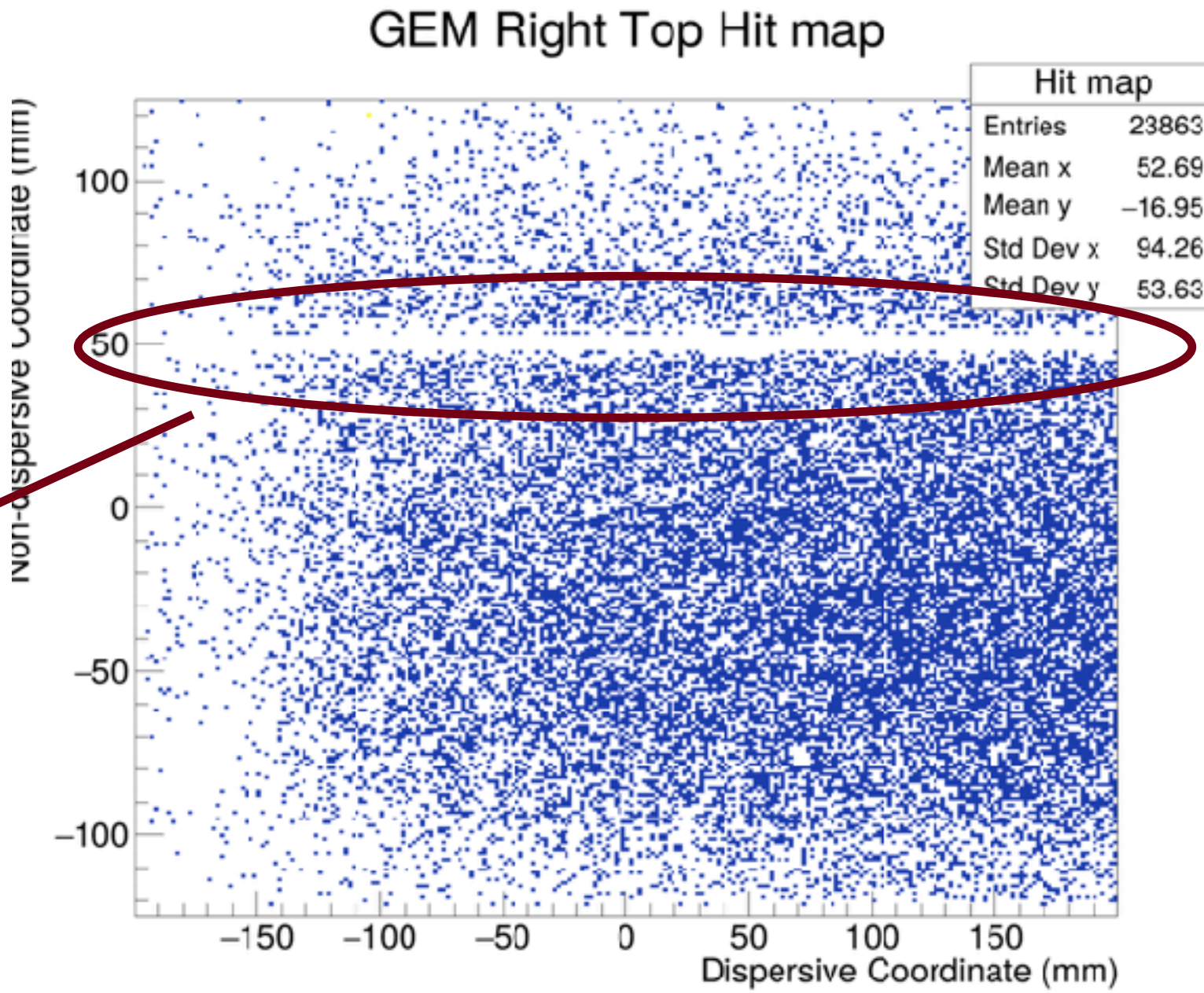
GEMs

- Run 2797 - cosmic run
- Supposedly “Full” readout, so left and right arms
  - For some reason few clusters on left GEMs (~2x less on Left Top, ~30x less on Left Bottom)
  - Set to 3950 V, but maybe APVs not configured?
- Right Top has known dead region due to APV



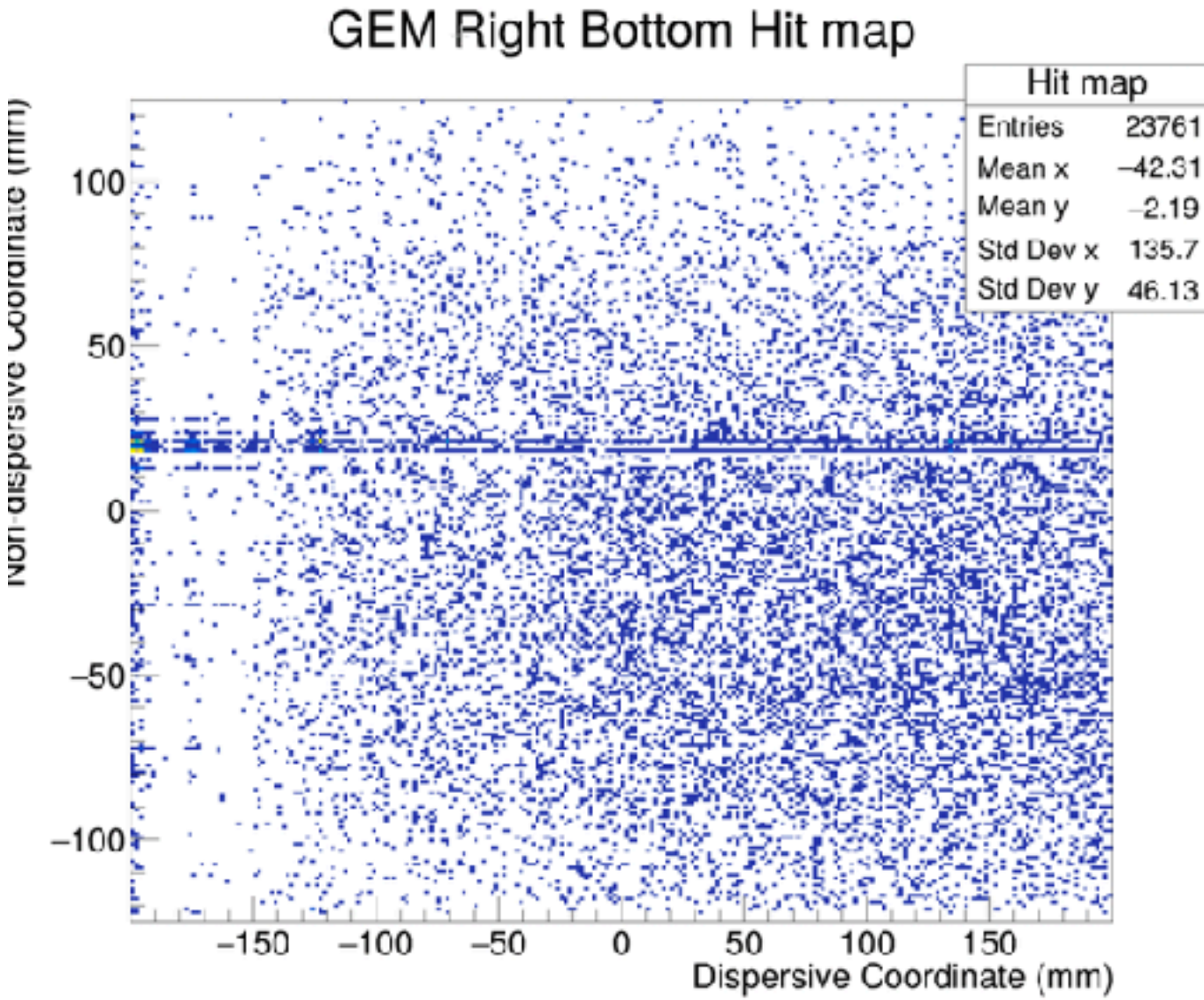
GEMs


Raw Waveform
ADC
Ch
rawwave
Entries 768
Mean x 64
Mean y 2.568
Std Dev x 37.32
Std Dev y 1.631

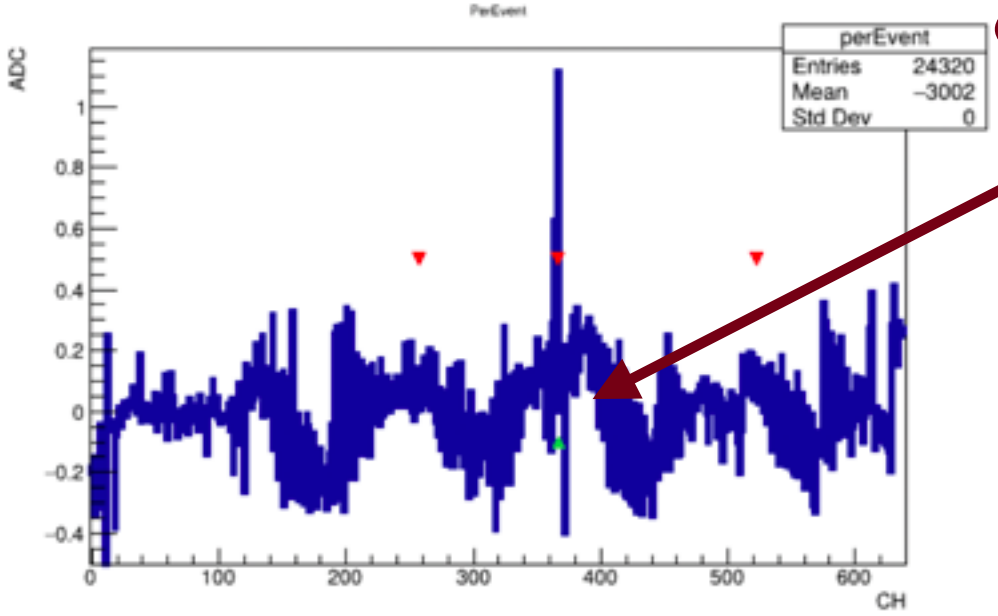

GEM Right Top Hit map
Hit map
Entries 23863
Mean x 52.69
Mean y −16.95
Std Dev x 94.26
Std Dev y 53.63
Non-dispersive Coordinate (mm)
Dispersive Coordinate (mm)

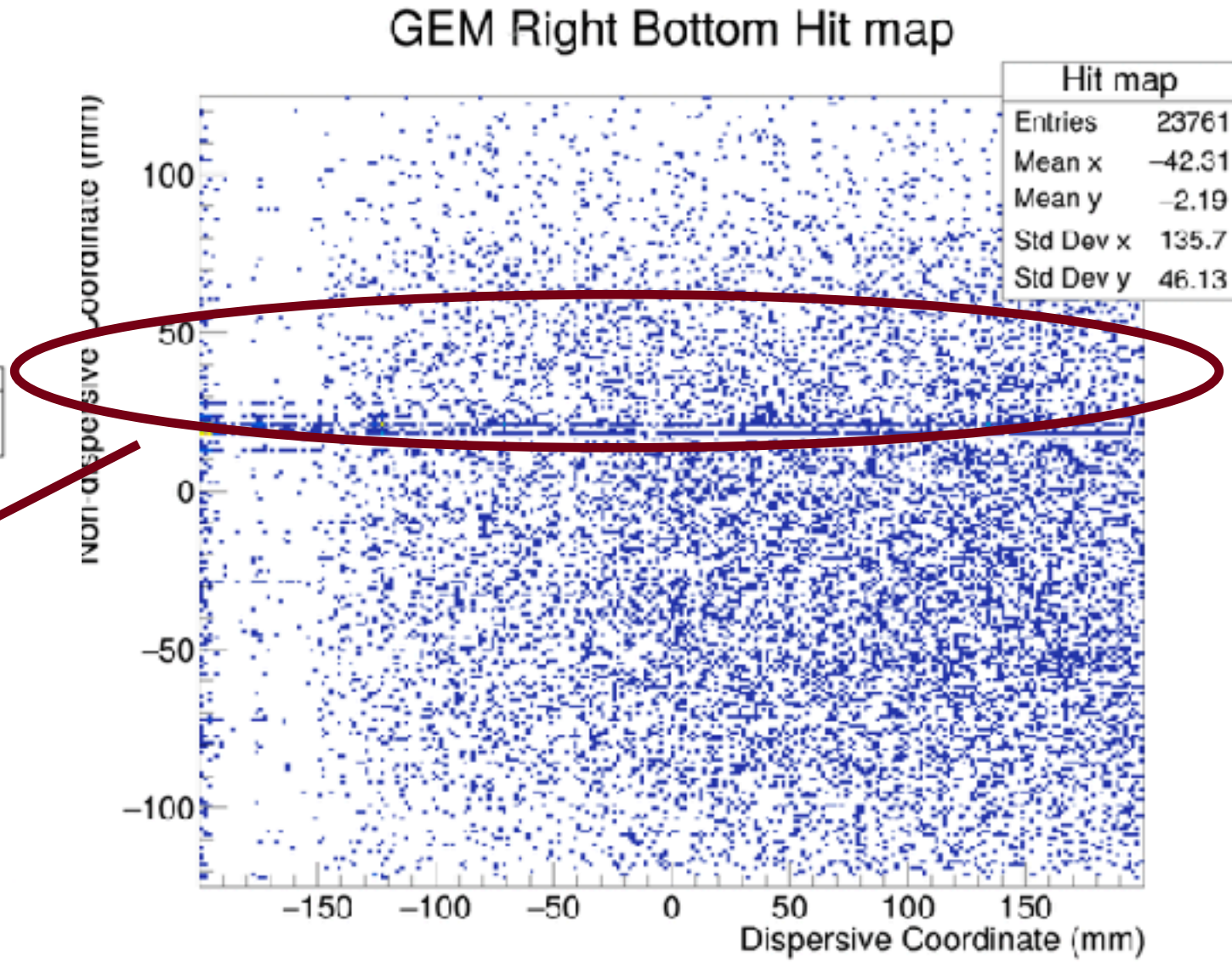
GEMs

- **Run 2797 - cosmic run**
- **Right bottom has noisy region, needs to be masked**

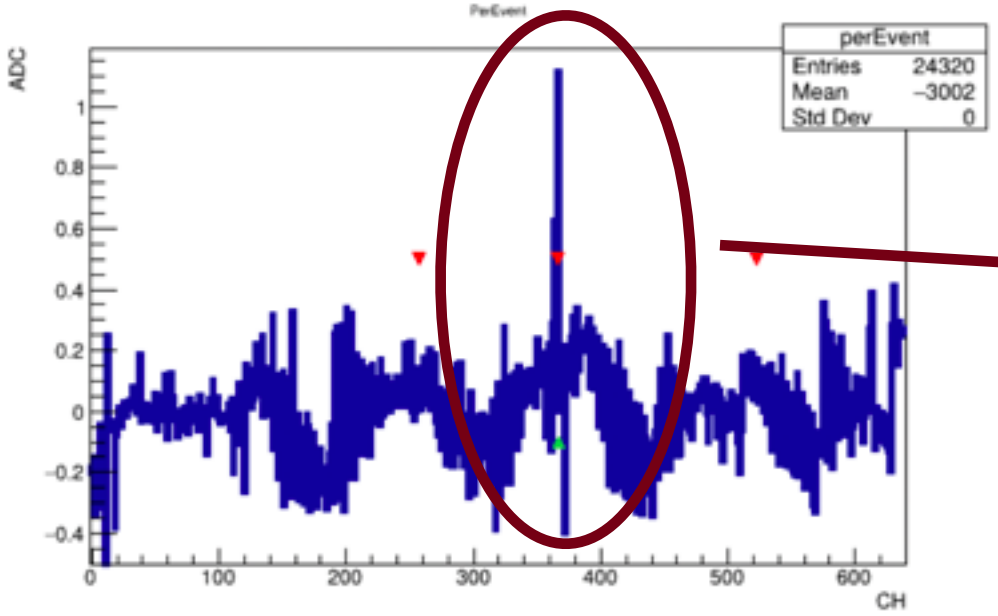


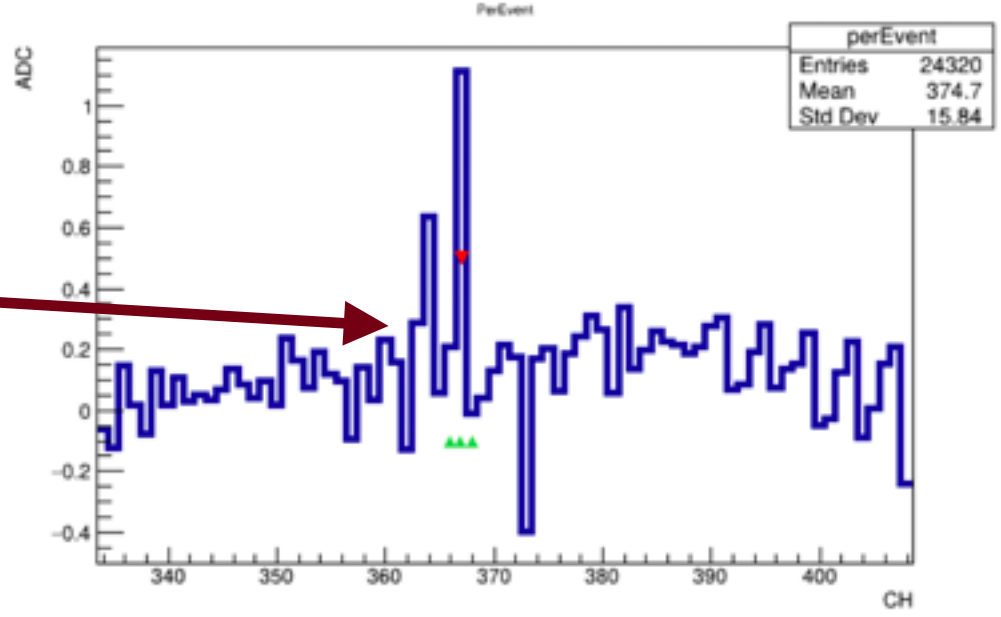
GEMs


PerEvent
perEvent
Entries 24320
Mean −3002
Std Dev 0
ADC
CH


GEM Right Bottom Hit map
Hit map
Entries 23761
Mean x −42.31
Mean y −2.19
Std Dev x 135.7
Std Dev y 46.13
Non dispersive Coordinate (mm)
Dispersive Coordinate (mm)

GEMs





Sometimes both peaks are identified as clusters

Next Steps

- **Identify accelerator conditions for run 2797**
- **Determine why Left GEMs see fewer clusters**